Name of the Student: _____ **Roll no:** _____

CLASS: - T.E (CSE) **Division:** **Course: - PCC305CSE- Machine Learning Lab**

Experiment No. 03

** Multiple Linear Regression Modeling**

Marks: /10

Date of Performance: / /2026 **Sign with Date:**

Aim: To build, evaluate, and interpret a Multiple Linear Regression (MLR) model using Python and Scikit-Learn.

Lab Objective: To assess its predictive performance using standard regression evaluation metrics (MSE, RMSE, and R2).

Problem Statement:

To build, evaluate, and interpret a Multiple Linear Regression (MLR) model using Python and Scikit-Learn on a real-world open-access dataset, and to assess its predictive performance using standard regression evaluation metrics (MSE, RMSE, and R^2).

Theory:

Multiple Linear Regression Equation

Multiple Linear Regression models the relationship between two or more independent variables (features, X_1, X_2, ……. X_p) and a continuous target variable (Y):

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots\ldots + \beta_p X_p + \epsilon$$

Where:

- β_0 is the intercept (value of Y when all $X_i = 0$).
- β_i are the regression coefficients representing the expected change in Y for a unit change in X_i, holding all other features constant.
- ϵ is the error term (residuals).

Evaluation Metrics:

- **Mean Squared Error (MSE):** Measures the average squared difference between actual (y_i) and predicted $\hat{y}_i$) values. Punishes larger errors heavily.

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

- **Root Mean Squared Error (RMSE):** The square root of MSE. Expresses error in the same physical units as the target variable Y.

$$RMSE = \sqrt{MSE}$$

- **Coefficient of Determination R^2 Score):** Measures the proportion of variance in the target variable explained by the model's feature set.

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

$R^2 = 1.0$: Perfect predictions.

$R^2 = 0.0$: Model performs no better than taking the mean of Y.

$R^2 < 0.0$: Model is worse than taking the mean baseline.

Dataset Selection:

For this lab, you may use the California Housing Dataset available directly via sklearn.datasets or download the Wine Quality Dataset / Bike Sharing Dataset from the UCI Machine Learning Repository.

Target Variable (Y): MedHouseVal (Median house value in 100,000s)

Features (X): MedInc (Median income), HouseAge, AveRooms, AveBedrms, Population, AveOccup, Latitude, Longitude.

Step-by-Step Execution

Step 1: Environment Setup & Data Ingestion:

Load core data science libraries and the dataset. Initialize Python dependencies (pandas, numpy, matplotlib, seaborn, sklearn) and load your dataset into memory.

Step 2: Exploratory Data Analysis & Preprocessing:

Inspect missing values, correlations, and scale features. Inspect data types, handle missing values (if any), check feature correlations, and divide the dataset into feature matrix X and target vector y.

Step 3: Model Instantiation & Training:

Fit the Multiple Linear Regression model. Instantiate Scikit-Learn's LinearRegression model and fit it onto the normalized training subset..

Step 4: Model Prediction & Evaluation:

Calculate MSE, RMSE, and R^2 scores. Generate predictions on the test set (X_test_scaled) and calculate standard metrics against y_test.

Step 5: Residual Analysis & Visualization:

Plot predicted values vs actual targets and examine residuals. Visualize how closely predictions line up with true values and analyze residual errors.

Algorithm: Multiple Linear Regression & Evaluation:

INPUT : Open-access dataset D = {X_1, X_2, ..., X_p, Y}

OUTPUT : Model Parameters (intercept β_0, coefficients β_1..p), Predictions ($\hat{y}$), Evaluation Metrics (MSE, RMSE, R^2)

STEP 1: [Data Ingestion & Inspection]

1.1 Load dataset D into memory (e.g., Pandas DataFrame).

1.2 Check for missing/null values and handle them (imputation or deletion).

1.3 Separate the dataset into feature matrix X (dimensions N x p) and target vector y (dimension N x 1).

STEP 2: [Dataset Splitting]

2.1 Split matrix X and vector y into training and testing subsets using ratio (e.g., 80:20):

(X_train, y_train), (X_test, y_test)

STEP 3: [Feature Scaling / Normalization]

3.1 Compute training mean μ_train and standard deviation σ_train for each feature in X_train.

3.2 Standardize training features:

X_train_scaled = (X_train - μ_train) / σ_train

3.3 Standardize testing features using training parameters:

X_test_scaled = (X_test - μ_train) / σ_train

STEP 4: [Model Construction & Training]

4.1 Prepend a column of ones to X_train_scaled to account for the intercept β_0.

4.2 Solve for weight parameters β using Ordinary Least Squares (OLS) closed-form solution:

$\beta = (X_train_scaled^T * X_train_scaled)^{-1} * X_train_scaled^T * y_train$

(OR compute via Gradient Descent iteratively until convergence).

STEP 5: [Prediction]

5.1 Generate target predictions $\hat{y}$_test for the unseen test features X_test_scaled:

$\hat{y}_test = \beta_0 + (\beta_1 * X_1) + (\beta_2 * X_2) + ... + (\beta_p * X_p)$

STEP 6: [Performance Evaluation]

6.1 Compute Mean Squared Error (MSE):

$MSE = (1 / N_{test}) * \Sigma (y_{test_i} - \hat{y}_{test_i})^2$

6.2 Compute Root Mean Squared Error (RMSE):

$RMSE = \sqrt{MSE}$

6.3 Compute Total Sum of Squares (SST) and Residual Sum of Squares (SSR):

$SST = \Sigma (y_{test_i} - \bar{y}_{test})^2$

$SSR = \Sigma (y_{test_i} - \hat{y}_{test_i})^2$

6.4 Compute R-Squared (R^2):

$R^2 = 1 - (SSR / SST)$

STEP 7: [Visualization & Reporting]

7.1 Plot Actual vs. Predicted values.

7.2 Plot Residuals ($y_{test} - \hat{y}_{test}$) against Predicted values ($\hat{y}_{test}$).

7.3 Output calculated β coefficients, MSE, RMSE, and R^2 scores.

Questions:

1. 1. What is the main difference between Simple Linear Regression and Multiple Linear Regression?
2. Why is feature scaling (like StandardScaler) important before fitting a Multiple Linear Regression model?
3. What does an R^2 score of 0.72 indicate about your model's predictions?
4. What is a "residual" in regression analysis, and what does a good residual plot look like.
5. Why is RMSE often preferred over MSE when reporting final performance to stakeholders?

Attendance and Timely Submission (10)	Execution (10)	Observations and Records (10)	Journal Documentation (10)	Oral (10)

Signature of Subject Teacher